Figure 9.25 Region Search on a k-d tree: returns all the points in a k-d tree within a given hyper-sphere. This means looking for points within a given Euclidean distance from the sphere's center. We start search from the root: it's not within the sphere. The point is on `A`'s left branch, so we need to traverse it; but even if `A` is not within the sphere, the split line through it intersects the sphere, so there is a portion of the sphere intersecting the right branch of `A` as well (highlighted in the top-left figure).
For the following steps, we are showing in parallel the execution on all branches at a given level, for the sake of space (this is also a good hint that this processes could be executed in parallel…).

Listing 9.11 shows the pseudo-code for this method: as you can see, it's pretty similar to the regular NN-search.

Listing 9.11 The pointsInSphere method

```
function pointsInSphere(node, center, radius)
                  #1
  if node == null
then                                                    #2
    return []
  else
    points ←
[]
   #3
    dist ← distance(node.point,
center)                            #4
     if dist < radius
then                                                     #5
```